

Datatype Mapping (SQL Server as source): SQL Server / CLR / CoreDataType / Oracle / PostgreSQL

SQL-Server-first schema / stored-proc migration reference — defaults mirror the TflOmniDb provider type mappers.

SQL Server	CLR / .NET	CoreDataType	Oracle	PostgreSQL	Notes
Integer & boolean					
BIT	bool	Boolean	NUMBER(1,0)	BOOLEAN	Oracle <23c: NUMBER(1) 0/1; 23c+ has BOOLEAN
TINYINT	byte	(none)	NUMBER(3,0)	SMALLINT	TINYINT is unsigned 0-255; no byte CoreDataType -> use Int16
SMALLINT	short / Int16	Int16	NUMBER(5,0)	SMALLINT	
INT	int / Int32	Int32	NUMBER(10,0)	INTEGER	
BIGINT	long / Int64	Int64	NUMBER(19,0)	BIGINT	
INT IDENTITY	long (identity)	(none)	GENERATED AS IDENTITY	GENERATED AS IDENTITY / serial	Identity = KeyGeneration strategy, not a type
Exact numeric					
DECIMAL(p,s)	decimal	Decimal	NUMBER(p,s)	NUMERIC(p,s)	NUMERIC == DECIMAL
NUMERIC(p,s)	decimal	Decimal	NUMBER(p,s)	NUMERIC(p,s)	Oracle NUMBER max precision 38
MONEY	decimal	Money	NUMBER(19,4)	NUMERIC(19,4)	Avoid PG MONEY (locale); canonical NUMERIC(19,4)
SMALLMONEY	decimal	(none)	NUMBER(10,4)	NUMERIC(10,4)	Map to Decimal/Money; range ~ +/-214,748.3647
Approximate numeric					
REAL	float / Single	Single	BINARY_FLOAT	REAL	REAL = FLOAT(24), 4-byte
FLOAT	double / Double	Double	BINARY_DOUBLE	DOUBLE PRECISION	FLOAT(53) = 8-byte double; FLOAT(24)->REAL
Character & text					
VARCHAR(n)	string	AnsiString	VARCHAR2(n CHAR)	VARCHAR(n)	Oracle VARCHAR2 <= 4000 bytes (32767 extended)
NVARCHAR(n)	string	UnicodeString	NVARCHAR2(n)	VARCHAR(n)	Unicode; PG is UTF-8 native (no N-prefix)
CHAR(n)	string	AnsiStringFixed	CHAR(n CHAR)	CHAR(n)	Fixed length, space-padded
NCHAR(n)	string	UnicodeStringFixed	NCHAR(n)	CHAR(n)	Unicode fixed length
VARCHAR(MAX)	string	Clob	CLOB	TEXT	Large ANSI text (> 8000 bytes)
NVARCHAR(MAX)	string	NClob	NCLOB	TEXT	Large Unicode text
TEXT (legacy)	string	Clob	CLOB	TEXT	Deprecated -> use VARCHAR(MAX)
NTEXT (legacy)	string	NClob	NCLOB	TEXT	Deprecated -> use NVARCHAR(MAX)
Date & time					
DATE	DateTime / DateOnly	Date	DATE	DATE	Oracle DATE INCLUDES time-of-day (to sec)

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TIME(p)	TimeSpan / TimeOnly	Time	INTERVAL DAY TO SECOND	TIME(p)	Oracle has no TIME-of-day type
DATETIME2(p)	DateTime	DateTime	TIMESTAMP(p)	TIMESTAMP(p)	Preferred date+time; p = frac-sec precision
DATETIME (legacy)	DateTime	DateTime	TIMESTAMP(3)	TIMESTAMP(3)	~3.33 ms precision; range 1753-9999
SMALLDATETIME	DateTime	DateTime	TIMESTAMP(0)	TIMESTAMP(0)	Minute precision; 1900-2079
DATETIMEOFFSET	DateTimeOffset	DateTimeOffset	TIMESTAMP WITH TIME ZONE	TIMESTAMPTZ	Timezone-aware
Binary					
VARBINARY(n)	byte[]	Binary	RAW(n)	BYTEA	Oracle RAW <= 2000 bytes; beyond -> BLOB
BINARY(n)	byte[]	BinaryFixed	RAW(n)	BYTEA	Fixed; PG has no fixed-length binary
VARBINARY(MAX)	byte[]	Blob	BLOB	BYTEA	Large binary object
IMAGE (legacy)	byte[]	Blob	BLOB	BYTEA	Deprecated -> use VARBINARY(MAX)
Identifier & concurrency					
UNIQUEIDENTIFIER	Guid	Guid	RAW(16)	UUID	Oracle SYS_GUID(); PG native UUID
ROWVERSION	byte[] (rowversion)	(none)	(none)	(xmin sys col)	IsRowVersion flag -> app-managed int version
XML & JSON					
XML	string / SqlXml	Xml	XMLTYPE	XML	Native XML on all three
JSON / NVARCHAR(MAX)	string	Json	JSON (21c+)/CLOB	JSONB	PG JSONB preferred; Oracle <21c CLOB + IS JSON
No direct equivalent (redesign)					
SQL_VARIANT	object	(none)	(none / ANYDATA)	(none)	Redesign to a typed column
HIERARCHYID	(tree)	(none)	(none)	LTREE (extension)	Remodel as a materialized path
GEOGRAPHY/GEOMETRY	(spatial)	(none)	SDO_GEOMETRY	geometry/geography	Oracle Spatial / PostGIS extension

General notes

1. Read left-to-right: this is a SQL-Server-FIRST reference. Find the SQL Server column/parameter type in the left column, then read across for the .NET/CLR type, the TfiOmniDb CoreDataType to use, and the Oracle / PostgreSQL equivalents when porting the schema.
2. Stored procedures: the CoreDataType column is what you pass to StoredProcedureCall.Input/Output(..., CoreDataType.X) and to [DbColumn(CoreType = ...)]. The DAL's per-provider type mappers translate it to the native types shown here.
3. Oracle SP-parameter caveat: a CoreDataType.DateTime parameter binds as DbType.DateTime (TIMESTAMP) on Oracle, NOT DateTime2 - Oracle's managed provider rejects DbType.DateTime2 ('Value does not fall within the expected range'). SQL Server / PostgreSQL keep full-precision DateTime2. (CoreTypeDbType.Map.)
4. '(none)' in the CoreDataType column marks a SQL Server type with no canonical enum value - map it to the nearest CoreDataType, or set a per-column native override via [DbColumn] SqlServerType / OracleType / PostgresType.
5. Money is deliberately NUMERIC(19,4) on all three engines - PostgreSQL's MONEY type is locale-dependent and unsafe for financial code.
6. Identifier casing: Oracle folds unquoted identifiers to UPPER, PostgreSQL to lower; SQL Server is case-insensitive. The DAL always quotes - call an unquoted-uppercase Oracle object by its UPPERCASE name.
7. Oracle DATE always carries a time component (to seconds). Use TIMESTAMP for fractional seconds; for pure dates, enforce truncation in the app.
8. Booleans: SQL Server BIT -> Oracle <23c NUMBER(1) 0/1 (23c+ native BOOLEAN) -> PostgreSQL native BOOLEAN.
9. Auto-increment: SQL Server IDENTITY -> PostgreSQL & Oracle 12c+ GENERATED ... AS IDENTITY; older Oracle needs a SEQUENCE + trigger.
10. Row versioning: there is no portable equivalent to SQL Server ROWVERSION. Use an app-managed integer version column (TfiOmniDb IsRowVersion) checked in the UPDATE WHERE clause; Oracle ORA_ROWSCN / PostgreSQL xmin are not drop-in replacements.

11. LOBs: SQL Server (N)VARCHAR(MAX)/VARBINARY(MAX) -> Oracle CLOB/NCLOB/BLOB -> PostgreSQL TEXT/BYTEA. PostgreSQL is UTF-8 native, so NVARCHAR and VARCHAR both collapse to VARCHAR/TEXT.